Intro to Data Science - HW 7

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```
# Enter your name here: Tim Soron
```

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```
# 1. I did this homework by myself, with help from the book and the professor.
```

In our last assignment, we explored data visualization in R using the **ggplot2** package. This homework continues to use ggplot2, but this time, with **maps**. Additionally, we will merge datasets using the built-in **merge()** function, which provides a similar capability to a JOIN in SQL (don't worry if you do not know SQL). Many analytical strategies require joining data from different sources based on a key - a field that two datasets have in common.

Step 1: Load the data

A. Some people may struggle to eat a healthy diet due to a lack of access to supermarkets, grocery stores, or other sources of healthy and affordable food. This limited availability to food is commonly referred to as **food insecurity**. However, as it is frequently linked to deep-rooted systemic injustices, many agree that the term "food apartheid" is more fitting.

There are numerous methods for measuring food store access and determining which areas are low-access neighborhoods with few healthy food options. In this homework assignment, we will focus on accessibility to sources of healthy food, as measured by the distance to a store in a given area.

The dataset available at https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv includes a variable labeled LAPOP1_10 that indicates the count of individuals residing more than 1 mile (for urban areas) or 10 miles (for rural areas) away from a supermarket.

To analyze the data, you can read it into a dataframe named **insecurity_df** and visually inspect its contents.

Please note in a comment what geographic area an observation (a row in the dataframe) represents - whether it pertains to a single city, a state, or another region.

Finally, use a command to display the number of observations and variables in insecurity_df.

```r
insecurity_df <- read.csv("https://data-science-intro.s3.us-east-2.amazonaws.com/FoodInsecurity.csv")
head(insecurity_df)
```

```
##     State         County Pop2010  LAPOP1_10 AveragePovertyRate
## 1 Alabama Autauga County   54571 18503.2255        16.13078591
## 2 Alabama Baldwin County  182265 45789.0465        11.84554563
## 3 Alabama Barbour County   27457  5634.3628        29.29932484
## 4 Alabama    Bibb County   22915   365.1407        12.19352439
## 5 Alabama  Blount County   57322  3902.2661        14.85074758
## 6 Alabama Bullock County   10914  7479.7500        30.65326247
##   MedianFamilyIncome  Largest_city             city_state abbr
## 1            69337.5    Prattville    Prattville, Alabama   AL
```

```
## 2        72665.74194        Daphne        Daphne, Alabama   AL
## 3        44792.44444       Eufaula       Eufaula, Alabama   AL
## 4            60645.5         Brent         Brent, Alabama   AL
## 5        60437.66667       Oneonta       Oneonta, Alabama   AL
## 6        39524.66667 Union Springs Union Springs, Alabama   AL
```

```
# Comment: Each row represents a county
```

```
# Comment: There are 3142 rows and 9 variables
```

B. Calculate the average of MedianFamilyIncome in insecurity_df. Explain in a comment why the mean() function might not work and how to fix it.

```
mean(as.numeric(insecurity_df$MedianFamilyIncome), na.rm = TRUE)
```

```
## Warning in mean(as.numeric(insecurity_df$MedianFamilyIncome), na.rm = TRUE):
## NAs introduced by coercion
```

```
## [1] 63973.97
```

```
# The mean function does not work because the column stores non-numeric values
# along with NA values. I fixed this by using as.numeric to change the type and
# used na.rm=TRUE to remove the NA values.
```

C. What is the population of the most populous county in the dataframe? Which state is it in?

```
insecurity_df[which.max(insecurity_df$Pop2010), ]
```

```
##          State             County Pop2010 LAPOP1_10 AveragePovertyRate
## 205 California Los Angeles County 9818605  641340.3        16.24263362
##     MedianFamilyIncome Largest_city              city_state abbr
## 205        79980.20418  Los Angeles Los Angeles, California   CA
```

```
# LA, California has the largest population at 9818605
```

D. Without a reference point, it is difficult to understand the significance of the values in **LAPOP1_10** (remember, this is the variable that shows the number of people living too far from a store and thus considered at risk of food insecurity). Create a new column in insecurity_df called **insecurity_ratio** that represents the ratio of LAPOP1_10 to Pop2010. Explain what it means in a comment.

```
insecurity_df$insecurity_ratio <- insecurity_df$LAPOP1_10/insecurity_df$Pop2010
# This column represents the percent of people in the population that are
# affected by food insecurity
```

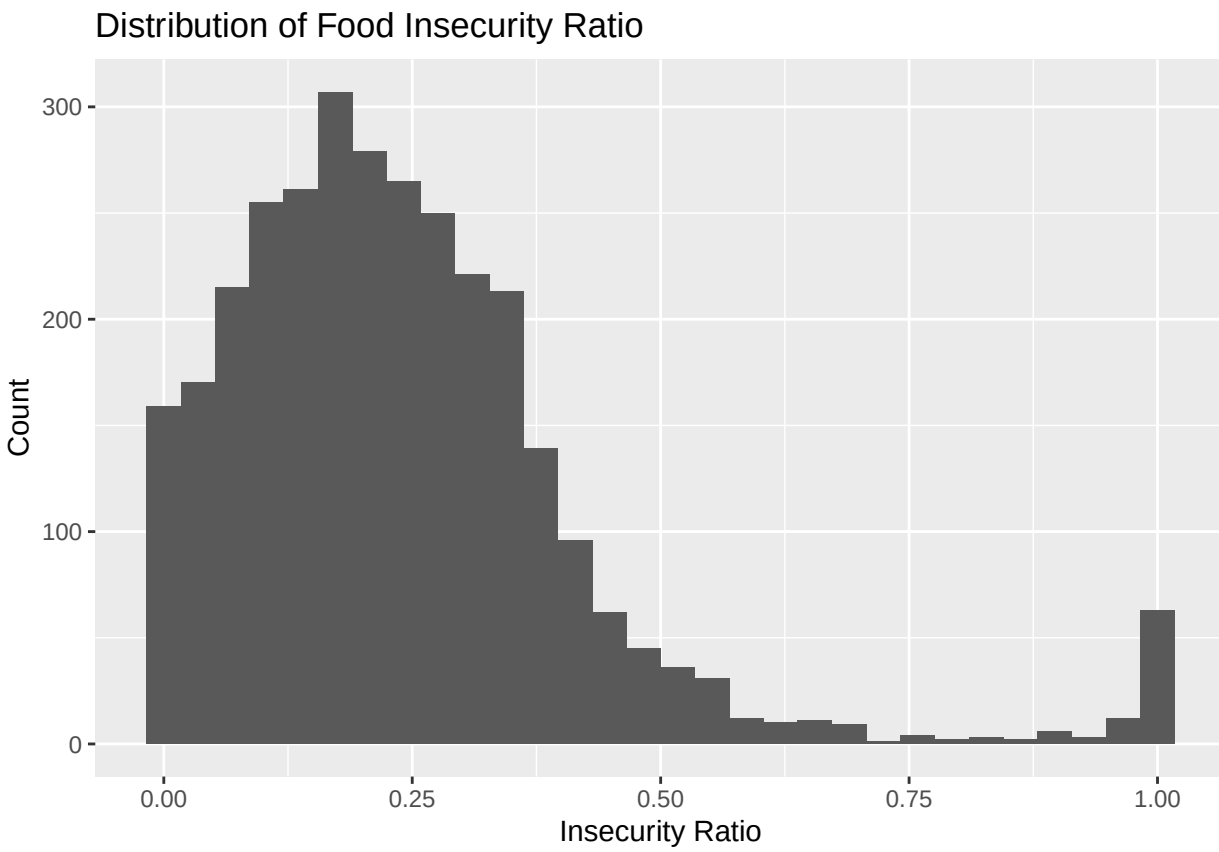
E. Provide descriptive statistics for the insecurity ratio (min, max, mean, and median) using a single function. Create a histogram with ggplot2 to visualize the distribution.

```
library(ggplot2)
summary(insecurity_df$insecurity_ratio)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.0000  0.1193  0.2147  0.2439  0.3237  1.0000
```

```
ggplot(insecurity_df, aes(x = insecurity_ratio)) + geom_histogram() + xlab("Insecurity Ratio") +
    ylab("Count") + ggtitle("Distribution of Food Insecurity Ratio")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Step 2: Merge the food apartheid data with the city data

A. Read the JSON file at https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json into a dataframe called **population_df**.

Use a function to display the variable data types in population_df and add comments explaining what each variable contains.

```
library(jsonlite)
population_df <- fromJSON("https://intro-datascience.s3.us-east-2.amazonaws.com/cities.json")
str(population_df)
```

```
## 'data.frame':    1000 obs. of  7 variables:
##  $ city                    : chr  "New York" "Los Angeles" "Chicago" "Houston" ...
##  $ growth_from_2000_to_2013: chr  "4.8%" "4.8%" "-6.1%" "11.0%" ...
##  $ latitude                : num  40.7 34.1 41.9 29.8 40 ...
##  $ longitude               : num  -74 -118.2 -87.6 -95.4 -75.2 ...
##  $ population              : chr  "8405837" "3884307" "2718782" "2195914" ...
##  $ rank                    : chr  "1" "2" "3" "4" ...
##  $ state                   : chr  "New York" "California" "Illinois" "Texas" ...
```

```
# city is the name of the city, growth_from_2000_to_2013 is the population
# growth from 2000-2013, latitude and longitude represent their names,
# population represents the cites population, rank is the cities rank based on
# population, and state is what state the city is in.
```

B. To merge insecurity_df and pop_df, we need a common column. We'll use the **city_state variable** in insecurity_df, which we'll create by joining the city and state columns in population_df. Explain the code provided:

```r
population_df$city_state <- paste0(population_df$city, ", ", population_df$state)
# Comment: This adds a new column to population_df called city_state, it joins
# the city and state separated by a comman before putting in the new column
```

C. Merge the two dataframes using the city_state column from both dataframes and store the resulting dataframe in **merged_df**.

```r
merged_df <- merge(insecurity_df, population_df, by = "city_state")
```

D. Display the dimensions (number of observations and variables) of merged_df and comment on the number of variables.

```r
dim(merged_df)
```

```
## [1] 423  17
```

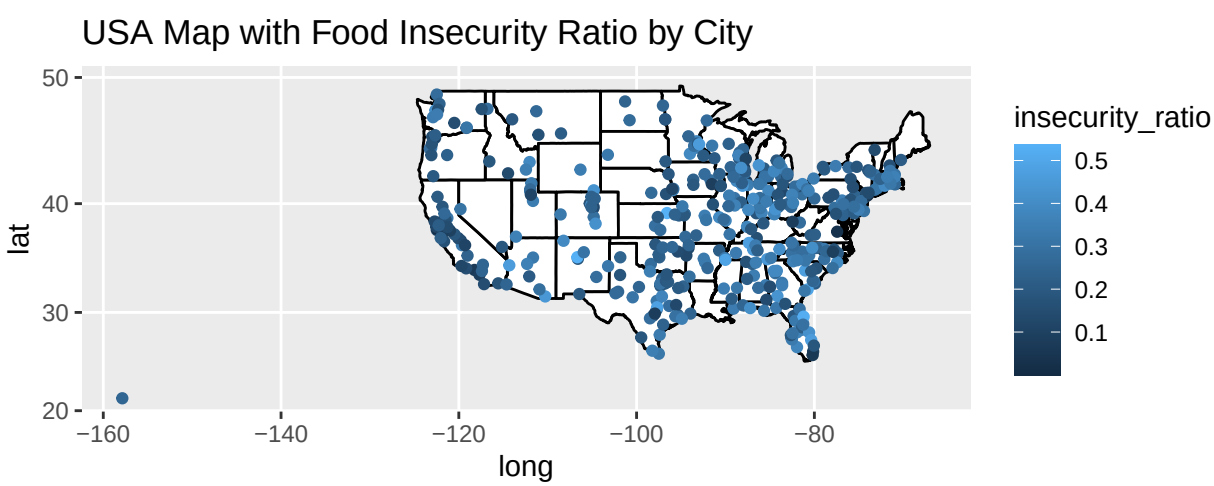
```r
# there are more variables this time becuase the dataframes have been merged
# and the all of the columns are now in one df
```

Step 3: Visualize the data

A. Plot points for each city in merged_df on a map of the United States. Use color to represent the insecurity ratio.

```r
library(ggplot2)
library(maps)

us <- map_data("state")
ggplot() + geom_polygon(data = us, aes(x = long, y = lat, group = group), fill = "white",
    color = "black") + expand_limits(x = us$long, y = us$lat) + coord_map() + geom_point(data = merged_
    aes(x = longitude, y = latitude, color = insecurity_ratio)) + ggtitle("USA Map with Food Insecurity
```



B. The map you just created is probably not perfect; comment briefly on what could be improved.

```r
# Comment: The map is small and not centered also there is a point not in the
# us. This could be improved by chaning the x and y limits along with removing
# the point
```

Step 4: Group by State

A. Create a dataframe of state-by-state average **insecurity_ratio** using **group_by()** and **summarize()**. Save the result in **state_insecurity_df**.

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
state_insecurity_df <- summarize(group_by(merged_df, state), avg_insecurity_ratio = mean(as.numeric(inse
    na.rm = TRUE)))
```

B. Identify the most and least food-insecure states in state_insecurity_df and their food insecurity ratios. Show the code you used to determine them.

```r
arrange(state_insecurity_df, avg_insecurity_ratio)[1, ]
```

```
## # A tibble: 1 x 2
##   state   avg_insecurity_ratio
##   <chr>                  <dbl>
## 1 Vermont                0.139
```

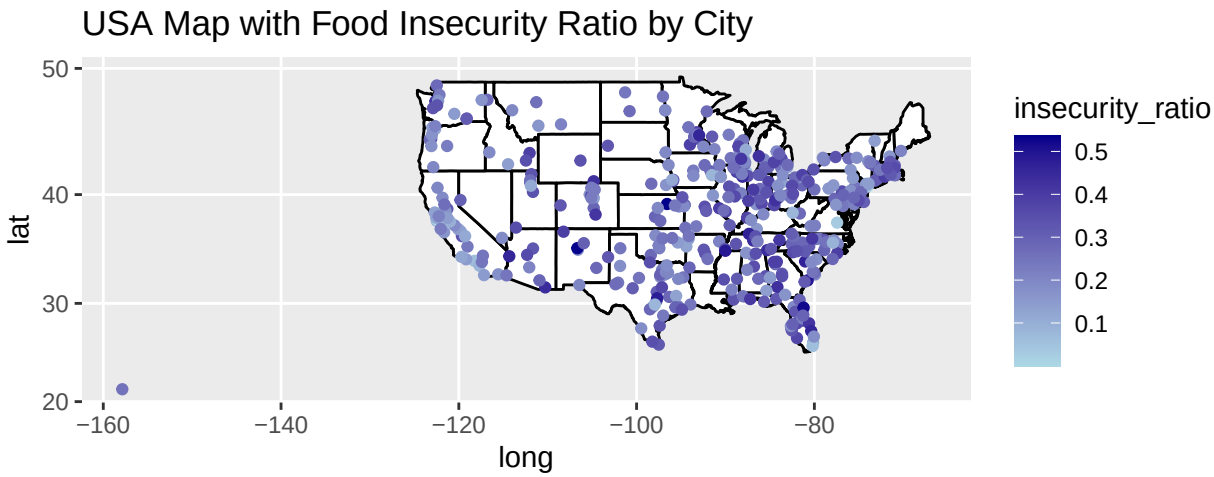
```r
arrange(state_insecurity_df, desc(avg_insecurity_ratio))[1, ]
```

```
## # A tibble: 1 x 2
##   state   avg_insecurity_ratio
##   <chr>                  <dbl>
## 1 Wyoming                0.382
```

Step 5: Create a map of the United States, with the color of each state representing the average food insecurity ratio in state_insecurity_df

A. Ensure you correctly set the map limits and use coord_map appropriately, and make sure that lighter shades represent lower food insecurity.

```r
ggplot() + geom_polygon(data = us, aes(x = long, y = lat, group = group), fill = "white",
    color = "black") + expand_limits(x = us$long, y = us$lat) + coord_map() + geom_point(data = merged_
    aes(x = longitude, y = latitude, color = insecurity_ratio)) + scale_color_gradient(low = "lightblue
    high = "darkblue") + ggtitle("USA Map with Food Insecurity Ratio by City")
```



Step 6: Data Analysis

A. Explain in a comment what insights you can glean about food insecurity from the maps you created in this homework.

```
# Comment: Food insecurity happens all over the US, but we can see that in the
# mid-west and the south have a higher percentage of food insecurity. This
# could be the fact that everything is more spaced out and there are less food
# stores, or it could be an economic indicator. The highest percenatge of a
# states population with food insecurity is Wyoming and the least is Vermont.
# We can also see that there are many more cities in the east and west coasts,
# because of this there are probably more food stores, which reduces the
# percentage of food insecurity.
```